CLASS TEST

CO2-AT2.

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87/15

① (a) Define Data mining:-

Data mining is the process of discovering useful patterns, relationships, and knowledges from large amounts of the data.

Two Basic Data mining Tasks:

① classification.

② clustering.

steps in KDD:-

① Data cleaning
② Data integration.
③ Data selection
④ Data transformation
⑤ Data mining.
⑥ Pattern Evaluation
⑦ knowledge presentation

(b) Data cleaning Data cleaning is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

Two Data quality issues:-

① missing values

② Noisy data.

Handling missing values:

→ Replace with mean / median / mode

Ex:- missing Age = Replace with average age.
incorrect value = smooth using binning.

2 ⓐ Data integration:-

Data integration combi... data from multiple sources into one un... dataset

challenges:-

① Data redundancy

② Data Inconsistency.

Handling Redundancy:-

-> Improve duplicate records.

-> correlation analysis.

Handling inconsistency:-

-> standardize formats

-> Resolve conflicting values

ⓑ

covariance Analysis:-

covariance analysis measures how two variables change together.

two covariance forms:-

① positive covariance.

② Negative covariance.

③ a) Given

A = 2,3,5,4,6

B = 5,8,10,11,14

Both A and B generally increase together

The covariance is positive, so the stock prices rise together, not independently.

(b) Architecture of Data mining system:-

Data sources
|
Data base /Data warehouse.
|
Data cleaning & Integration.
|
Data mining Engine
|
pattern Evaluation.
|
Knowledge Base.
|
user Interface.

components:-
-> Data base / Data warehouse - stores data.
-> Data cleaning - Removes errors.
-> Data mining Engine - Finds patterns.
-> pattern Evaluation - selects useful pattern.

(4) a) Given
$x = \{4, 5, 6, 7, 9\}$ $y = \{8, 3, 7, 5, 6\}$

mean $(x) = 6.2$ mean$(y) = 5.8$

$$cov(x, y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{n}$$

x	y	$(x - 6.2)$	$(y - 5.8)$	product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56.

sum = -1.80

covariance

= -1.80/5

$\boxed{= -0.36}$

(b) major steps of Data preprocessing:-
① Data cleaning ② Data integration ③ Data Trans-formation ④ Data Reduction ⑤ Data Discretization

pipeline:-
missing values -> Remove outliers -> standardize formats -> Normalize Data -> Reduce Data -> Discretize Data.

⑤ (a)
Data set:
13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70

Equal - frequency
Bin: 1 13, 15, 16, 16, 19, 20, 20, 21, 22

mean = 17.
smoothed : 17, 17, 17, 17, 17, 17, 17, 17

Bin 2 22, 25, 25, 25, 25, 30, 33, 33, 35

mean = 28.
smoothed: 28, 28, 28, 28, 28, 28, 28, 28, 28

Bin 3 35, 35, 35, 36, 40, 45, 46, 52, 70.

mean = 43
smoothed: 43, 43, 43, 43, 43, 43, 43, 43, 43

(b)

Data Discretization it converts continuous values into intervals (bins).

Importance:-

-> Reduces data size.

-> Removes noise

-> improves mining accuracy.

Equal width	Equal Frequency
① Equal interval size	① Equal number of records
② simple to implement	② Better for uneven data.
③ may create empty bins	③ Balanced distribution.

Ex:- Equal width : 0-20, 21-40, 41-60

Equal frequency : Each bin contains the same number of values.

smooth using binning.